

Franziska Hoerbst

bouncing between research, art, activism & science communication

📍 Norwich, UK

✳ 1998, Austria

✉ franziska@hoerbst.net

🌐 linktree

Research and education

since 2024

Postdoc

● John Innes Centre

Computational Systems Biology

Statistical pitfalls in the sciences and what we can do to resolve them

2020 – 2024

PhD

● John Innes Centre & University of East Anglia

Computational Systems Biology

Doctoral Thesis supervised by Richard Morris
Bayesian Inference in the Analysis of RNA-Seq Data

2017 – 2020

BSc

● University of Vienna

Molecular Biology

Thesis in Computational Biophysics with Bojan Zagrovic at Max Perutz Labs

Exploring the Physicochemical Foundation behind Travelling Autologous Protein-mRNA Pairs in the Phloem of Plants

Science communication & teaching

2020 – 2022

🌐 more here

● 5MinutenClimateChance

Climate science podcast & radio broadcast

Co-founder, host, journalistic researcher & producer

→ Austrian SDG Award 2020

→ Austrian Environmental Journalism Prize 2020

→ Top 5 Salto Awards 2021

2022

Norwich, UK

● University teaching

Mathematics for Computer Scientists

Teaching assistant

2018 – 2020

Vienna, AT

● ScienceCenter Network Vienna

Hands-on science communication

in a job role called 'explainer', which is essentially a workshop host, who is developing hands-on and tinkering activities for audiences of all ages and backgrounds

is fascinated by

Life

Philosophy of science

Colours

Information Theory

Climate & Biodiversity Crises

Bayesian Statistics

The importance of cultural practices

Complex Systems

Slowing down & degrowth

Equality & Justice

The post carbon world

Nature & Humanity

New things every day

Research publications

2022	● <u>Exact Bayesian inference for the detection of graft-mobile transcripts from sequencing data</u>
Royal Society Interface	F Hoerbst, M Tomkins, S Gupta, F Apelt, J Kehr, F Kragler, RJ Morris
2023	● <u>baymobil: A Python package for detection of graft-mobile mRNA using exact Bayesian inference on RNA-Seq data</u>
Preprint	F Hoerbst, RJ Morris, M Tomkins
2024	● <u>A Closed-Form Solution to the 2-Sample Problem for Quantifying Changes in Gene Expression using Bayes Factors</u>
Preprint	F Hoerbst, GS Sidhu, M Tomkins, RJ Morris
2024	● <u>Re-analysis of mobile mRNA datasets highlights challenges in the detection of mobile transcripts from short-read RNA-Seq data</u>
Nature	P Paajanen, M Tomkins, F Hoerbst, R Veevers, M Heeney, HR Thomas, F Apelt, E Saplaoura, S Gupta, M Frank, DJ Walther, C Faulkner, J Kehr, F Kragler, RJ Morris
2024	● <u>Bayesian Inference in the Analysis of RNA-Seq Data</u>
Thesis	F Hoerbst
2025	● <u>Mass screening for mobile mRNAs</u>
In press	F Hoerbst, M Tomkins, RJ Morris
2024	● <u>A Bayesian framework for ranking genes based on their statistical evidence for differential expression</u>
Preprint	F Hoerbst, GS Sidhu, T Omori, M Tomkins, RJ Morris
2024	● <u>What is a differentially expressed gene?</u>
Preprint	F Hoerbst, GS Sidhu, M Tomkins, RJ Morris

Further publications & engagement

2024	● <u>Let's 'do different': Planetary Citizen Education Proposal</u>
Working paper	E Honeybun-Arnolda, P Stephanides, F Hoerbst, N Longhurst, D Maguire, H Hoechner
since 2020	● <u>5MinutenClimateChance</u>
Podcast	100+ episodes filled with snack-bites of climate research
2021	● <u>A day trip through phloem</u>
Blog	Easy read research project description
2023	● <u>Life out there? Scanning plant space for new communication networks</u>
Blog	Easy read paper summary
2023	● <u>Activism and Advocacy as a Researcher</u>
Panel	Panel discussion at the Annual Assembly of the Tyndall Centre for Climate Change Research

finds balance in

- Music and silence
- Nature and the outdoors
- Arts and crafts
- Tech and low-tech
- Exploring movement and movements

speaks

- German and English
- Python fluently
- Cada día un pocito más español

loves speaking

- at international scientific conferences
- on panels (like previously on Women in STEM, [climate activism as researchers](#), ...)
- at public engagement events (like e.g. [Pint of science](#))
- with kind humans around a dinner table

the miscellaneous

2024

→ **Best Presentation Award at MaxEnt24**
Maximum Entropy (MaxEnt) is a valued, historical conference series in the field of Bayesian statistics where researchers from quantum physics and philosophy to economy come together once a year since to learn from each other about probability theory

2023

JIC Director Award
A yearly award given away by Director Graham Moore to individuals to acknowledge and honour outstanding scientific as well as cultural contributions to the John Innes Centre

since 2022

🌐 more here

• **UEA Biodiversity and Climate Action Network**
Particularly involved in the Education Working group, with the vision of changing all curricula towards engaging all students in concepts surrounding global citizenship

since 2022

• **JIC Student Voice**
Running campaigns to support the John Innes student community to thrive

2021

• **Finalist DigiEduHack 2021**
Climate Change Digital Education Hackathon

from 2012

• **Radio Orange 94.0, Freies Radio Freistadt, & Radius 106.6**
Speaker and producer

→ Student award by the Austrian Federal Ministry of Education and OE1 for best broadcast 2016

2017 & 2018

Coordination and execution of the charity radio marathons
4 Tage für den Sozialmarkt, 4 Tage für die Lebenshilfe

2016

→ Regional Movie Award @YOUKI Film Festival

2008 – 2017

Several accolades for music (Let's discuss the concept of music competitions, especially for kids and teenagers)

Contact for references

• Professor Richard J. Morris

Group leader at John Innes Centre

Primary PhD supervisor

Richard.Morris@jic.ac.uk

• Dr Noel Longhurst

Lecturer in Energy & Climate Change, School of Environmental Sciences, UEA and founding member of UEA BCAN

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